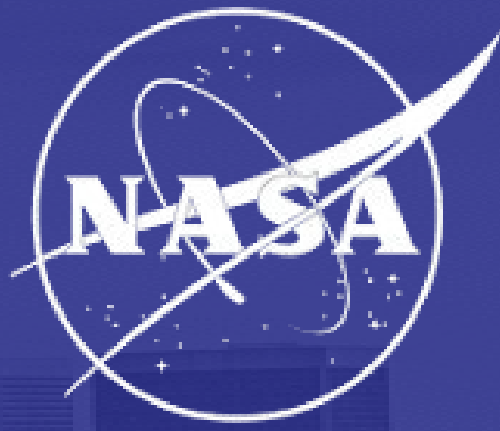


Quantifying Traffic-NO₂ Relationships Using EPA Air Quality Data and FHWA Traffic Counts



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ABSTRACT

Nitrogen dioxide (NO₂) is a trace gaseous pollutant formed when nitrogen and oxygen react under high-temperature conditions. This reaction occurs primarily during high-temperature combustion in vehicle engines and naturally during lightning strikes. Because urban areas experience heavy traffic, vehicles have become a major source of NO₂ emissions. NO₂ also contributes to the formation of harmful secondary pollutants. NO₂ reduces air quality, irritates the respiratory system, and is recognized as a key criteria pollutant. Monitoring NO₂ concentrations is essential for understanding environmental impacts and raising awareness of this public health concern. This study analyzes the relationship between traffic activity and NO₂ levels in New York City. Using hourly EPA NO₂ measurements and FHWA traffic counts from 2019–2021, we found a strong linear relationship during the morning peak hours, with the correlation weakening later in the day.

INTRODUCTION & METHODS

NO₂ is a reddish-brown gas produced mainly from high-temperature combustion, especially in urban traffic. Because it contributes to respiratory irritation, degraded air quality, and acid rain, understanding how NO₂ varies with daily human activity is important for major cities. Traffic remains one of the primary sources, making it useful to examine how vehicle patterns influence surface NO₂ levels.

Hourly NO₂ surface concentrations were collected from the Environmental Protection Agency (EPA) reference-grade monitoring stations across New York City for the years 2019–2021. These measurements, reported in parts per billion (ppb), provide reliable ground-level observations. Hourly traffic activity was obtained from the Federal Highway Administration (FHWA) vehicle-count sensors located along interstate highways, including tolling stations and automated monitoring sites.

Both datasets were time-matched and screened for missing or inconsistent values. After aligning the NO₂ and traffic datasets by hour, we performed statistical correlation analysis to evaluate how changes in vehicle volume relate to changes in NO₂ concentrations.

RESULTS

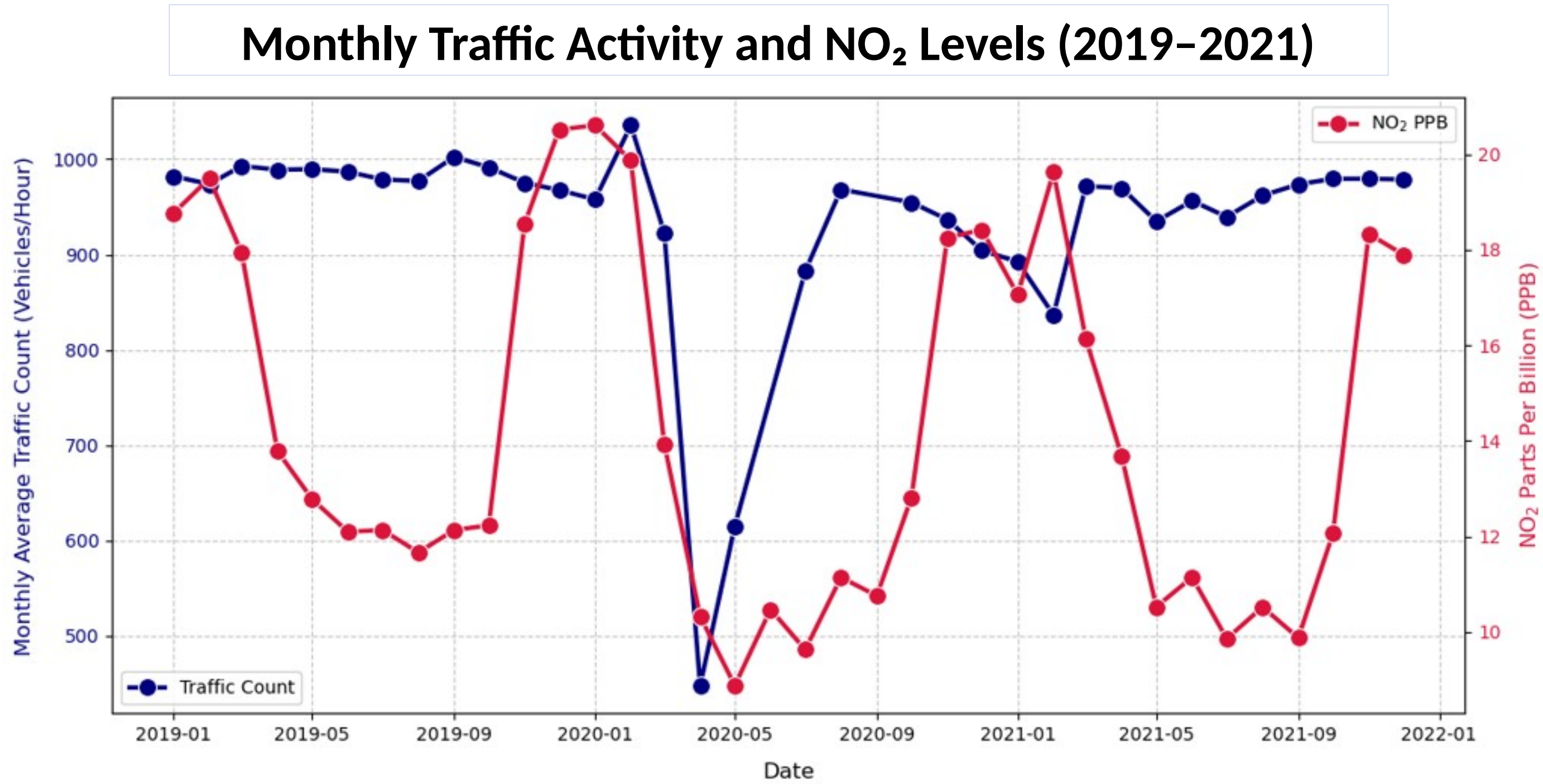


Figure 2: Monthly average traffic counts (blue, left axis) and surface NO₂ concentrations (red, right axis) from 2019 to 2021. The sharp decrease in early 2020 reflects reduced road activity and emissions during COVID-19 restrictions, followed by gradual recovery through 2021.

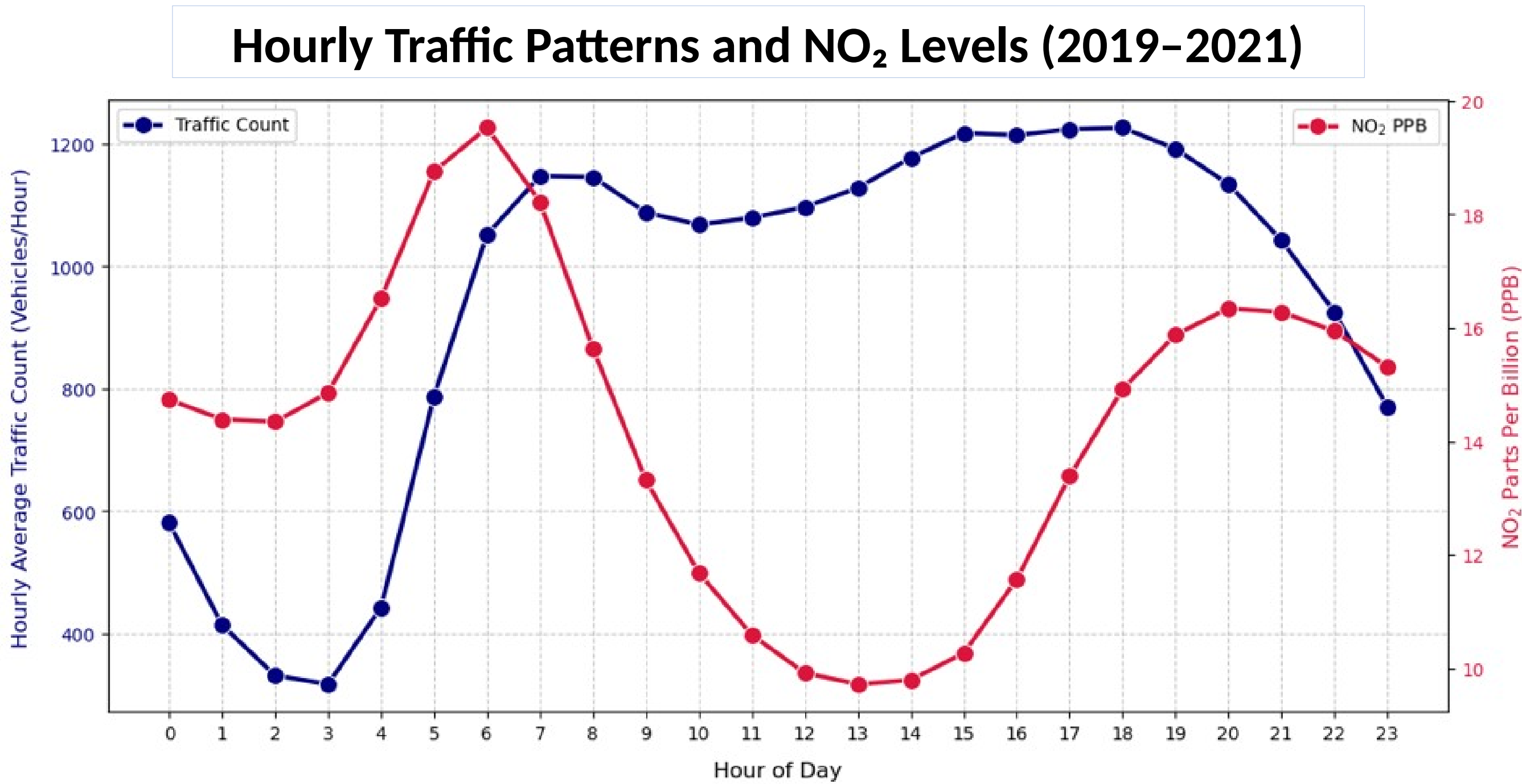


Figure 3: Diurnal variation of hourly traffic counts (blue, left axis) and surface NO₂ concentrations (red, right axis). Values represent hourly averages over the study period for matched EPA and FHWA stations in New York City.

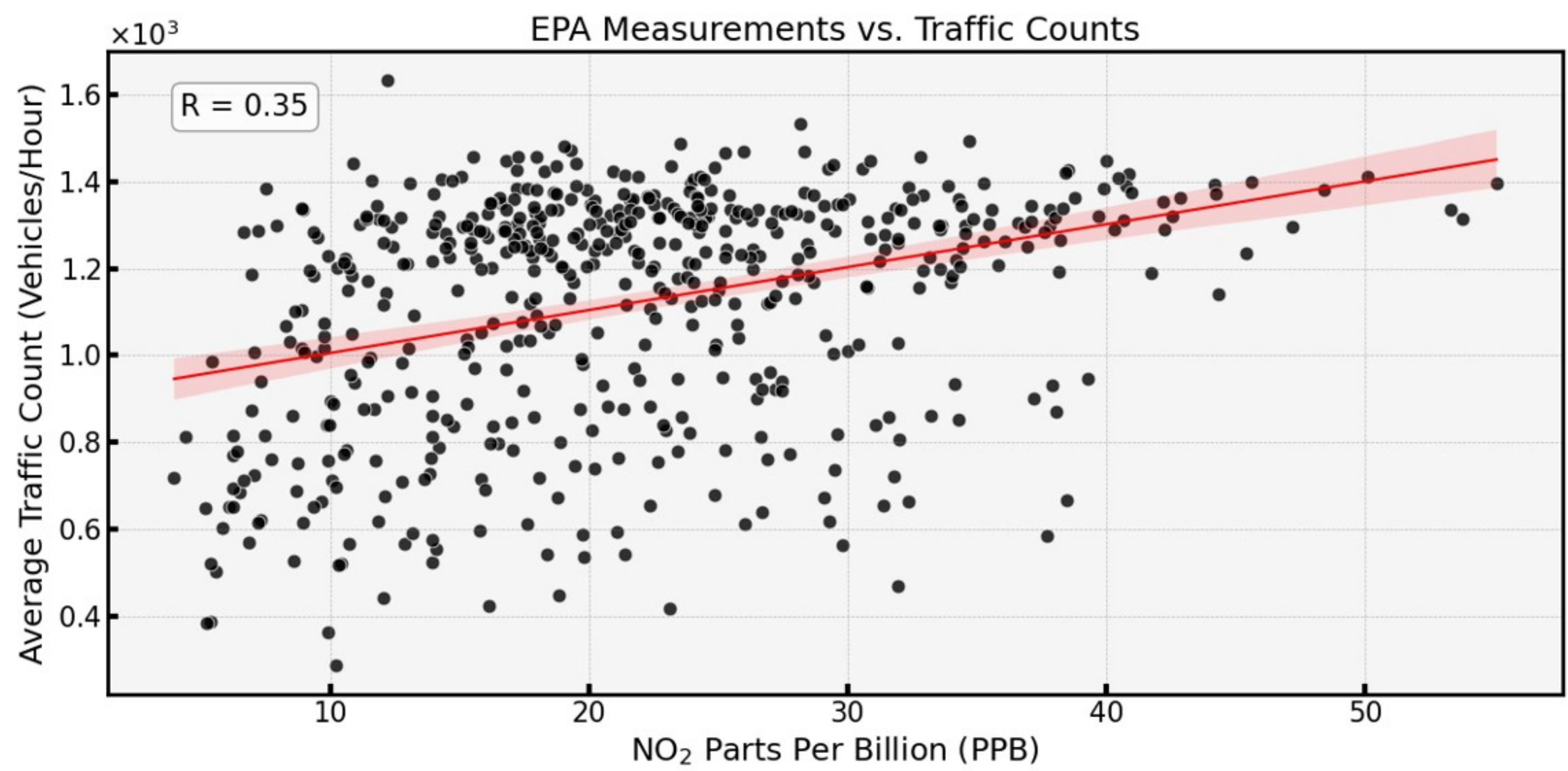


Figure 4: EPA hourly NO₂ surface concentrations (ppb) plotted against FHWA hourly traffic volume (vehicles/hour). The positive regression slope and correlation coefficient ($R = 0.35$) indicate that increases in traffic activity are associated with elevated NO₂ levels. Dispersion in the data reflects the influence of meteorology, boundary-layer depth, and photochemical processes throughout the day.

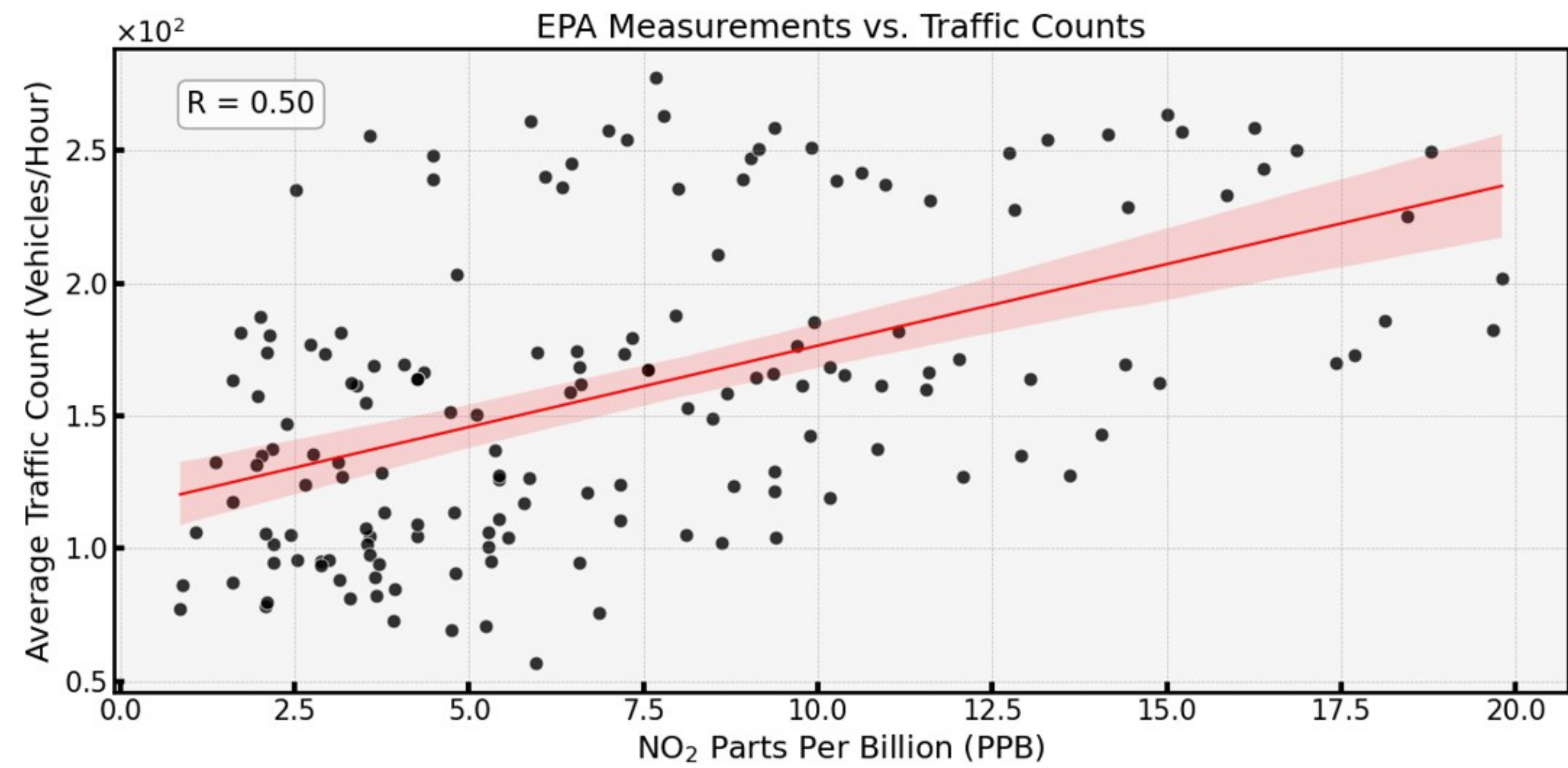


Figure 5: Correlation between EPA hourly NO₂ measurements and FHWA hourly traffic volume for an isolated station pair with minimal surrounding confounding sources. The regression line shows a clear positive trend ($R = 0.50$), demonstrating that closely paired stations reveal a stronger traffic–NO₂ relationship.

Published Traffic–Pollutant Correlations for Comparison With This Study

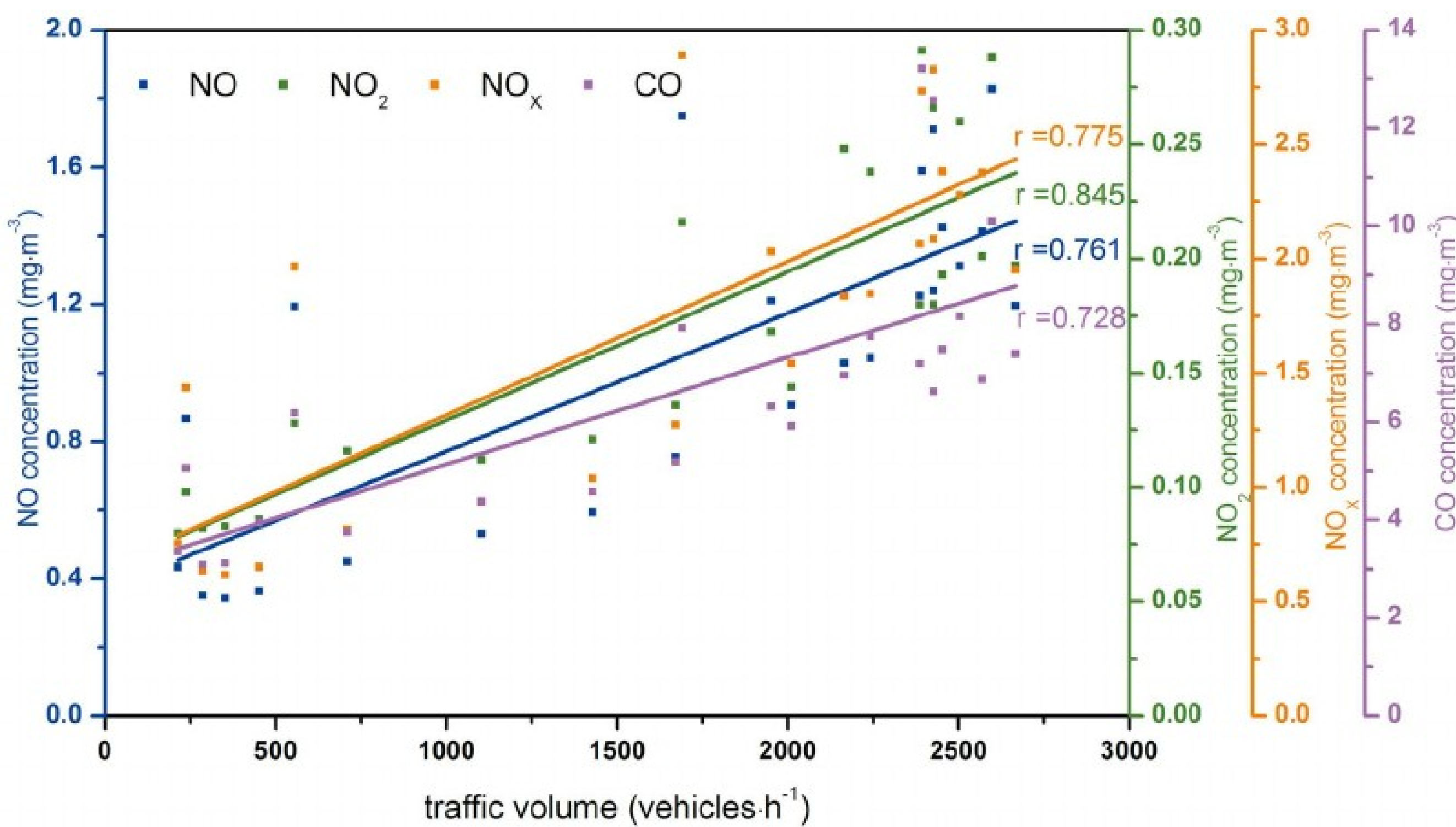


Figure 6: High correlations reported in a controlled road-tunnel study from Shanghai, China. Tunnel conditions reduce atmospheric mixing, resulting in stronger traffic–pollutant relationships than those observed in our open-environment data.

CONCLUSIONS

- Traffic activity shows a clear influence on surface NO₂ levels, with the strongest alignment occurring during morning peak hours when atmospheric mixing is limited.
- Monthly trends from 2019–2021 reveal that large reductions in traffic during the COVID-19 shutdowns corresponded with substantial declines in NO₂ concentrations in New York City.
- Correlations between EPA NO₂ and FHWA traffic data are moderate in open-air environments ($R \approx 0.35$ – 0.50) because meteorological conditions, boundary-layer depth, and photochemistry dilute and redistribute emissions throughout the day.
- Comparison with a controlled road-tunnel study in Shanghai ($r = 0.73$ – 0.85) demonstrates that enclosed environments produce much stronger traffic–pollutant correlations, highlighting the greater complexity and variability in real-world urban atmospheric conditions.

CITATIONS

Gyawali *et al.*, (2022), Tracking NO₂ Pollution Changes Over Texas: Synthesis of In Situ and Satellite Observations, JGR, <https://doi.org/10.1029/2022JD037473>
Zhou *et al.*, (2014) Study on the Traffic Air Pollution inside and outside a Road Tunnel in Shanghai, China, <https://doi.org/10.1371/journal.pone.0112195>

Figure 1: EPA and FHWA Stations in New York

